Michael Vinciguerra

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in AI-driven threat detection, machine learning model development, and Ethernet protocol testing and analysis. Built production-grade security systems integrating Splunk SIEM, custom SOAR workflows, and LLMs; trained and fine-tuned machine learning models, achieving high accuracy on real-world datasets. Proven collaborator with engineers from Cisco, Intel, AMD, and Apple, with a strong foundation in cybersecurity operations, Python automation, and network infrastructure.

EDUCATION

University of New Hampshire — *Durham, NH*

Bachelor of Science in Computer Science — *May 2026*

Relevant Coursework: Machine Learning, NLP, Cloud Computing, Cybersecurity Principles, Operating Systems, Data Structures & Algorithms, Database Systems & Design, Object-Oriented Design, Concurrency

EXPERIENCE

Network and Interoperability Engineer — *Durham, NH*

University of New Hampshire Interoperability Labs (IOL) — *February 2024 – June 2026*

- Led IEEE 802.3 compliance and interoperability testing with global technology partners including AMD, Apple, Intel, Cisco, HP, and Eero, with focused expertise in MAC layer testing (Clause 4) and Flow Control (Clause 31); configured and validated networking switches against standard requirements.
- Conducted network protocol analysis using Wireshark, executed troubleshooting and validation across Layer 2 and Layer 3 environments, and performed throughput and performance analysis with Spirent and Ixia.
- Acted as primary point of contact during live vendor testing sessions with engineers from AMD, Apple, Intel, Cisco, HP, and Eero; diagnosed protocol failures in real time, communicated findings across technical and non-technical stakeholders, and led post-session review meetings to present test results and remediation paths.
- Built a Python SNMP rack management system replacing a legacy C# application, featuring a multi-threaded Streamlit interface enabling real-time simultaneous monitoring and control of 12+ Tripp Lite PDUs by all lab technicians.
- Developed a Python automation tool for device selection and formatted Word/Excel report generation, cutting lab setup time by 40%.
- Coordinated testing logistics as Reservation Manager for 1+ year across 20+ global vendor engagements; managed communications, device specification collection, and live on-call troubleshooting with vendor engineers, maintaining zero missed testing windows across all sessions.
- Mentored 4+ new hire technicians on IEEE standards, lab equipment, and debugging methodology, enabling all four to independently lead client testing sessions within their first month.

AI Trainer, Software Engineering / Coding Domain — *Remote*

Handshake AI Fellowship — *June 2026 - Present*

- Evaluated and scored responses from frontier large language models (LLMs) on multi-turn software engineering and coding tasks, assessing response quality, helpfulness, and communication.
- Applied a structured evaluation rubric to rate model completions on a 1-7 scale across dimensions including helpfulness, technical clarity, tone, response structure, and adaptability to user needs.
- Authored detailed written rationales for each score and conducted comparative preference evaluations across multiple candidate responses to the same coding prompt.
- Maintained high inter-rater agreement and scoring consistency, passing a rigorous calibration screening before contributing to production data.

- Generated high-quality human feedback data used to improve the reasoning, communication, and overall response quality of AI coding assistants.

PROJECTS

Threat Detection & Analysis Learning Model

- Architected an AI-driven autonomous security analyst integrating Splunk SIEM for live event ingestion, a custom SOAR workflow for automated threat classification, and Ollama LLMs for structured incident response report generation, extending SOC coverage to 24/7.
- Built a retrieval-augmented generation (RAG) pipeline with vector database integration for context-aware threat hunting during active investigations; benchmarked multiple LLMs and SOAR platforms for model deployment in production; presented at UNH CS Poster Symposium, Spring 2026.

Urban Heat Island Predictive Modeling

- Trained a Random Forest regression model (R^2=0.747, RMSE=2.55°C) on a 19,800-point spatial grid using Landsat 8 thermal imagery, NYC building footprints, and street tree census data; managed the full ML pipeline from feature engineering through deployment.
- Simulated three climate warming scenarios and modeled green infrastructure interventions, achieving 1.28-1.90°C cooling reductions; identified impervious surfaces, NDVI, and vegetation score as top predictors.

Steam Review Sentiment Analysis

- Collected and processed 555K+ real-world Steam game reviews across 37 titles via the public API; built and compared multiple machine learning models for binary sentiment classification using Python and scikit-learn.
- Fine-tuned a BERT transformer model using HuggingFace Transformers, achieving over 99% classification accuracy and significantly outperforming all classical models, with the strongest gains on negative reviews.

ACTIVITIES & LEADERSHIP

Pi Kappa Phi *Founding Father & Risk Manager*

Co-founded the chapter from the ground up, growing to 80+ active members; authored bylaws, organizational structure, and recruitment frameworks still in use today. Developed and enforced chapter compliance policies as Risk Manager and coordinated logistics for a formal chapter event. Mentored members on academic and professional development; contributed to organizing fundraising campaigns raising thousands for The Ability Experience.

SKILLS

Languages: Python, Java, C, SQL

Security: SIEM (Splunk), Incident Response, Threat Hunting, Microsoft Defender, CrowdStrike, SOAR Workflows, Log Analysis, Security Monitoring, Incident Analysis

Machine Learning: scikit-learn, BERT, HuggingFace Transformers, RAG, Model Development

Networking & Dev: Layer 2/3 Networking, Network Validation, Network Troubleshooting, TCP/IP, Ethernet, VLANs, Wireshark, IEEE 802.3, Spirent, Ixia, Git